Data Base System Lab 4
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IN-LAB TASK

1.

Worksheet | Query Builder

```
SELECT
    FIRST_NAME,LAST_NAME
    SALARY
    SALARY + (SALARY * 0.10) - (SALARY * 0.05) AS new_salary
FROM EMPLOYEES;
```

Script Output × | Query Result ×

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	SALARY	NEW_SALARY
1	Steven	King	25200
2	Neena	Kochhar	17850
3	Lex	De Haan	17850
4	Alexander	Hunold	9450
5	Bruce	Ernst	6300
6	David	Austin	5040
7	Valli	Pataballa	5040
8	Diana	Lorentz	4410
9	Nancy	Greenberg	12608.4
10	Daniel	Faviet	9450
11	John	Chen	8610
12	Ismael	Sciarra	8085
13	Jose Manuel	Urman	8190
14	Luis	Popp	7245

2.

```
SELECT
    e.FIRST_NAME,
    e.LAST_NAME,
    e.SALARY,
    e.DEPARTMENT_ID
FROM EMPLOYEES e
JOIN DEPARTMENTS d
 ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
WHERE d.DEPARTMENT_NAME IN ('IT', 'Finance')
AND e.SALARY BETWEEN 60000 AND 100000;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 0 in 0.001 seconds

FIRST_NA...	LAST_NAME	SALARY	DEPARTM...

3.

```sql
SELECT
    FIRST_NAME,LAST_NAME,
    SALARY,
    DEPARTMENT_ID
FROM EMPLOYEES e
WHERE SALARY > (
    SELECT AVG(SALARY)
    FROM EMPLOYEES
    WHERE DEPARTMENT_ID = e.DEPARTMENT_ID
);
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 38 in 0.034 seconds

	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_ID
1	Steven	King	24000	90
2	Alexander	Hunold	9000	60
3	Bruce	Ernst	6000	60
4	Nancy	Greenberg	12008	100
5	Daniel	Faviet	9000	100
6	Den	Raphaely	11000	30
7	Matthew	Weiss	8000	50
8	Adam	Fripp	8200	50
9	Payam	Kaufling	7900	50
10	Shanta	Vollman	6500	50
11	Kevin	Mourgos	5800	50
12	Renske	Ladwig	3600	50
13	Trenna	Rajs	3500	50

4.

```sql
SELECT
    FIRST_NAME ,LAST_NAME,
    DEPARTMENT_ID,
    HIRE_DATE,
    SALARY
FROM EMPLOYEES
ORDER BY
    DEPARTMENT_ID ASC,
    HIRE_DATE DESC;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	HIRE_DATE	SALARY
1	Jennifer	Whalen	10	17-SEP-03	4400
2	Pat	Fay	20	17-AUG-05	6000
3	Michael	Hartstein	20	17-FEB-04	13000
4	Karen	Colmenares	30	10-AUG-07	2500
5	Guy	Himuro	30	15-NOV-06	2600
6	Shelli	Baida	30	24-DEC-05	2900
7	Sigal	Tobias	30	24-JUL-05	2800
8	Alexander	Khoo	30	18-MAY-03	3100
9	Den	Raphaely	30	07-DEC-02	11000
10	Susan	Mavris	40	07-JUN-02	6500
11	Steven	Markle	50	08-MAR-08	2200

5.

```sql
SELECT
    FIRST_NAME ,LAST_NAME,
    HIRE_DATE,
    ROUND((SYSDATE - HIRE_DATE) / 365, 2) AS experience_years
FROM EMPLOYEES;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	HIRE_DATE	EXPERIENCE_YEARS
1	Steven	King	17-JUN-03	22.67
2	Neena	Kochhar	21-SEP-05	20.4
3	Lex	De Haan	13-JAN-01	25.09
4	Alexander	Hunold	03-JAN-06	20.12
5	Bruce	Ernst	21-MAY-07	18.74
6	David	Austin	25-JUN-05	20.64
7	Valli	Pataballa	05-FEB-06	20.03
8	Diana	Lorentz	07-FEB-07	19.02

6.

```sql
FROM (
    SELECT
        FIRST_NAME, LAST_NAME,
        DEPARTMENT_ID,
        SALARY,
        ROW_NUMBER() OVER (PARTITION BY DEPARTMENT_ID ORDER BY SALARY DESC) AS rn
    FROM EMPLOYEES
)
WHERE rn <= 5
ORDER BY DEPARTMENT_ID, SALARY DESC;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 36 in 0.038 seconds

	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	SALARY	RN
1	Jennifer	Whalen	10	4400	1
2	Michael	Hartstein	20	13000	1
3	Pat	Fay	20	6000	2
4	Den	Raphaely	30	11000	1
5	Alexander	Khoo	30	3100	2
6	Shelli	Baida	30	2900	3
7	Sigal	Tobias	30	2800	4
8	Guy	Himuro	30	2600	5
9	Susan	Mavris	40	6500	1
10	Adam	Fripp	50	8200	1
11	Matthew	Weiss	50	8000	2
12	Payam	Kaufling	50	7900	3
13	Shanta	Vollman	50	6500	4
14	Kevin	Mourgos	50	5800	5

7.

```sql
SELECT CONCAT(CONCAT(FIRST_NAME, ' '), LAST_NAME) AS employee_name
FROM EMPLOYEES;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	EMPLOYEE_NAME
1	Ellen Abel
2	Sundar Ande
3	Mozhe Atkinson
4	David Austin
5	Hermann Baer
6	Shelli Baida
7	Amit Banda
8	Elizabeth Bates
9	Sarah Bell
10	David Bernstein
11	Laura Bissot
12	Harrison Bloom
13	Alexis Bull
14	Anthony Cabrio

8.

```sql
SELECT
    FIRST_NAME ,LAST_NAME,
    JOB_ID,
    SALARY,
    HIRE_DATE
FROM EMPLOYEES
WHERE HIRE_DATE < DATE '2015-01-01'
AND (
    (JOB_ID LIKE '%MAN%' AND SALARY <= 80000)
    OR
    (JOB_ID NOT LIKE '%MAN%' AND SALARY > 80000)
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 12 in 0.019 seconds

	FIRST_NAME	LAST_NAME	JOB_ID	SALARY	HIRE_DATE
1	Den	Raphaely	PU_MAN	11000	07-DEC-02
2	Matthew	Weiss	ST_MAN	8000	18-JUL-04
3	Adam	Fripp	ST_MAN	8200	10-APR-05
4	Payam	Kaufling	ST_MAN	7900	01-MAY-03
5	Shanta	Vollman	ST_MAN	6500	10-OCT-05
6	Kevin	Mourgos	ST_MAN	5800	16-NOV-07
7	John	Russell	SA_MAN	14000	01-OCT-04
8	Karen	Partners	SA_MAN	13500	05-JAN-05
9	Alberto	Errazuriz	SA_MAN	12000	10-MAR-05
10	Gerald	Cambrault	SA_MAN	11000	15-OCT-07
11	Eleni	Zlotkey	SA_MAN	10500	29-JAN-08
12	Michael	Hartstein	MK_MAN	13000	17-FEB-04

9.

```sql
SELECT
    UPPER(FIRST_NAME),UPPER(LAST_NAME),
    LENGTH(FIRST_NAME), length(LAST_NAME) AS name_length
FROM EMPLOYEES
WHERE DEPARTMENT_ID IN (
    SELECT DEPARTMENT_ID
    FROM EMPLOYEES
    GROUP BY DEPARTMENT_ID
    HAVING AVG(SALARY) > 70000
);
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 0 in 0.021 seconds

UPPER(FI...	UPPER(LA...	LENGTH(F...	NAME_LE...

10.

```sql
SELECT
    FIRST_NAME ,LAST_NAME
    SALARY AS base_salary,
    SALARY * 1.10 AS total_income
FROM EMPLOYEES;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	BASE_SALARY	TOTAL_INCOME
1	Steven	King	24000	26400
2	Neena	Kochhar	17000	18700
3	Lex	De Haan	17000	18700
4	Alexander	Hunold	9000	9900
5	Bruce	Ernst	6000	6600
6	David	Austin	4800	5280
7	Valli	Pataballa	4800	5280
8	Diana	Lorentz	4200	4620
9	Nancy	Greenberg	12008	13208.8
10	Daniel	Faviet	9000	9900
11	John	Chen	8200	9020
12	Ismael	Sciarra	7700	8470
13	Jose Manuel	Urman	7800	8580
14	Luis	Popp	6900	7590
15	Den	Raphaely	11000	12100
16	Alexander	Khoo	3100	3410
17	Shelli	Baida	2900	3190
18	Sigal	Tobias	2800	3080
19	Guy	Himuro	2600	2860

POST-LAB TASK

11.

```sql
SELECT
    FIRST_NAME , LAST_NAME,
    SALARY,
    ROUND((SALARY / (SELECT SUM(SALARY) FROM EMPLOYEES)) * 100, 2) AS salary_percentage
FROM EMPLOYEES
ORDER BY salary_percentage DESC;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	SALARY	SALARY_PERCENTAGE
1	Steven	King	24000	3.47
2	Neena	Kochhar	17000	2.46
3	Lex	De Haan	17000	2.46
4	John	Russell	14000	2.02
5	Karen	Partners	13500	1.95
6	Michael	Hartstein	13000	1.88
7	Nancy	Greenberg	12008	1.74
8	Alberto	Errazuriz	12000	1.74
9	Shelley	Higgins	12008	1.74
10	Lisa	Ozer	11500	1.66
11	Ellen	Abel	11000	1.59
12	Gerald	Cambrault	11000	1.59
13	Den	Raphaely	11000	1.59
14	Eleni	Zlotkey	10500	1.52
15	Clara	Vishney	10500	1.52
16	Janette	King	10000	1.45

12.

```sql
SELECT
    DEPARTMENT_NAME,
    UPPER(SUBSTR(DEPARTMENT_NAME, 1, 3)) AS dept_code
FROM DEPARTMENTS;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 27 in 0.002 seconds

	DEPARTMENT_NAME	DEPT_CODE
1	Administration	ADM
2	Marketing	MAR
3	Purchasing	PUR
4	Human Resources	HUM
5	Shipping	SHI
6	IT	IT
7	Public Relations	PUB
8	Sales	SAL
9	Executive	EXE
10	Finance	FIN
11	Accounting	ACC
12	Treasury	TRE
13	Corporate Tax	COR
14	Control And Credit	CON
15	Shareholder Services	SHA
16	Benefits	BEN
17	Manufacturing	MAN
18	Construction	CON
19	Contracting	CON
20	Operations	OPE
21	IT Support	IT
22	NOC	NOC
23	IT Helpdesk	IT
24	Government Sales	GOV
25	Retail Sales	RET
26	Recruiting	REC
27	Payroll	PAY

13.

Worksheet | Query Builder

```sql
SELECT
    PRODUCT_NAME,
    PRICE,
    QUANTITY,
    (PRICE * QUANTITY) - (PRICE * QUANTITY * 0.10) AS revised_price
FROM PRODUCTS;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 0 in 0.002 seconds

PRODUCT...	PRICE	QUANTITY	REVISED_...

14.

```sql
SELECT
    DEPARTMENT_ID,
    ROUND(AVG(SALARY), 2) AS avg_salary
FROM EMPLOYEES
GROUP BY DEPARTMENT_ID
ORDER BY DEPARTMENT_ID;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 12 in 0.002 seconds

	DEPARTMENT_ID	AVG_SALARY
1	10	4400
2	20	9500
3	30	4150
4	40	6500
5	50	3475.56
6	60	5760
7	70	10000
8	80	8955.88
9	90	19333.33
10	100	8601.33
11	110	10154
12	(null)	7000

15.

```sql
SELECT
    FIRST_NAME ,LAST_NAME ,
    NVL(TO_CHAR(COMMISSION_PCT), 'No Commission') AS commission
FROM EMPLOYEES;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	FIRST_NAME	LAST_NAME	COMMISSION
1	Steven	King	No Commission
2	Neena	Kochhar	No Commission
3	Lex	De Haan	No Commission
4	Alexander	Hunold	No Commission
5	Bruce	Ernst	No Commission
6	David	Austin	No Commission
7	Valli	Pataballa	No Commission
8	Diana	Lorentz	No Commission
9	Nancy	Greenberg	No Commission
10	Daniel	Faviet	No Commission
11	John	Chen	No Commission
12	Ismael	Sciarra	No Commission
13	Jose Manuel	Urman	No Commission
14	Luis	Popp	No Commission
15	Den	Raphaely	No Commission
16	Alexander	Khoo	No Commission

16.

```sql
SELECT
    UPPER(FIRST_NAME || ' ' || LAST_NAME) AS employee_name_upper,
    SUBSTR(JOB_ID, 1, 4) AS job_code,
    TO_CHAR(HIRE_DATE, 'DD-MON-YYYY') AS hire_date_formatted
FROM EMPLOYEES;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	EMPLOYEE_NAME_UPPER	JOB_CODE	HIRE_DATE_FORMATTED
1	STEVEN KING	AD_P	17-JUN-2003
2	NEENA KOCHHAR	AD_V	21-SEP-2005
3	LEX DE HAAN	AD_V	13-JAN-2001
4	ALEXANDER HUNOLD	IT_P	03-JAN-2006
5	BRUCE ERNST	IT_P	21-MAY-2007
6	DAVID AUSTIN	IT_P	25-JUN-2005
7	VALLI PATABALLA	IT_P	05-FEB-2006
8	DIANA LORENTZ	IT_P	07-FEB-2007
9	NANCY GREENBERG	FI_M	17-AUG-2002
10	DANIEL FAVIET	FI_A	16-AUG-2002
11	JOHN CHEN	FI_A	28-SEP-2005
12	ISMAEL SCIARRA	FI_A	30-SEP-2005
13	JOSE MANUEL URMAN	FI_A	07-MAR-2006
14	LUIS POPP	FI_A	07-DEC-2007

17.

```sql
SELECT
    FIRST_NAME || ' ' || LAST_NAME AS employee_name,
    SALARY,
    HIRE_DATE
FROM EMPLOYEES
WHERE
    (FIRST_NAME LIKE '%n' OR FIRST_NAME LIKE '%a%')
    AND SALARY < 80000
    AND HIRE_DATE BETWEEN DATE '2012-01-01' AND DATE '2022-12-31';
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 0 in 0.002 seconds

EMPLOYE...	SALARY	HIRE_DATE

18.

```sql
SELECT
    FIRST_NAME || ' ' || LAST_NAME AS employee_name,
    SALARY AS monthly_salary,
    SALARY * 12 AS annual_income
FROM EMPLOYEES;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | Fetched 50 rows in 0.002 seconds

	EMPLOYEE_NAME	MONTHLY_SALARY	ANNUAL_INCOME
1	Steven King	24000	288000
2	Neena Kochhar	17000	204000
3	Lex De Haan	17000	204000
4	Alexander Hunold	9000	108000
5	Bruce Ernst	6000	72000
6	David Austin	4800	57600
7	Valli Pataballa	4800	57600
8	Diana Lorentz	4200	50400
9	Nancy Greenberg	12008	144096
10	Daniel Faviet	9000	108000
11	John Chen	8200	98400

19.

```sql
SELECT
    FIRST_NAME,LAST_NAME,
    SALARY,
    HIRE_DATE
FROM EMPLOYEES
WHERE
    (FIRST_NAME LIKE 'A%' OR FIRST_NAME LIKE 'M%')
    AND SALARY > 70000
    AND HIRE_DATE BETWEEN DATE '2010-01-01' AND DATE '2020-12-31';
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 0 in 0.002 seconds

FIRST_NA...	LAST_NAME	SALARY	HIRE_DATE

20.

```sql
SELECT
    e.FIRST_NAME , e.LAST_NAME,
    d.DEPARTMENT_NAME,
    ROUND((SYSDATE - e.HIRE_DATE) / 365, 2) AS tenure_years,
    e.SALARY
FROM EMPLOYEES e
JOIN DEPARTMENTS d
    ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
WHERE
    (SYSDATE - e.HIRE_DATE) / 365 > 5
    AND e.SALARY > 60000
ORDER BY tenure_years DESC;
```

Script Output × | Query Result × | Query Result 1 ×

SQL | All Rows Fetched: 0 in 0.003 seconds

FIRST_NA...	LAST_NAME	DEPARTM...	TENURE_...	SALARY